Neural Morphogen Operators: Learning Continuous Reaction–Diffusion Dynamics from Spatial Transcriptomics

Anonymous Author(s)
Affiliation
Address
email

Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, an inductive bias that ties the learned function to one sampling lattice and leaves expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla\cdot(\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, and renders $\mathbf{D}_\theta$ readable as a diffusion length. Across 17 tissue sections from four technologies (198 runs), NMO improves masked-region reconstruction over graph, Gaussian-process and implicit neural-field baselines, beating every one with Holm-corrected $p \leq 0.003$ in paired tests over sections. Because twelve of those sections are serial sections of one brain, the conservative unit of analysis is the specimen, and at that level the separation from the *continuous* baselines is what the data establish most firmly: NMO wins on all 5 independent specimens with $d_z = 2.37$, the strongest outcome this design admits. Two controls localize the gain. An exactly parameter-matched ablation—the 0.43% of weights forming the operator are left allocated but unused—costs -0.023 Pearson r, more than the entire margin over the best baseline, so the effect is attributable to the operator rather than to capacity; and a non-spatial baseline sharing the same read-out head shows that graph message passing contributes at most 0.001, against 0.016 for NMO. Replacing the exponential integrator with explicit Euler costs $32\times$ more steps to reach the same tolerance, so the numerical machinery is load-bearing rather than decorative, and we prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, which delimits what any method of this class can identify from static tissue. Two limits are worth stating plainly. Against the *graph* baselines the specimen-level separation is small and statistically unresolved ($d_z = 0.32 - -0.81$ over 5 specimens, where the smallest attainable Wilcoxon p is 0.06), so more independent tissues, not more sections of one brain, is what would settle it; and the predicted fields are smoother than the measurements, on which NMO trails every baseline tested.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction–diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Morphogen Operators** (NMO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction–diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

Contributions.

1. We introduce a PDE-constrained neural operator for spatial omics, coupling a graph kernel-integral encoder, a spectral reaction–diffusion operator and a coordinate-free continuous decoder (Section 2).
2. We show that the resulting integrator is unconditionally stable and second-order accurate, with structurally guaranteed positive-definite diffusion and a uniform absorbing set for the field (Propositions **??**–**??**).
3. We show that the conventional physics-informed residual is uninformative for an exactly integrated operator, and derive the steady-state constraint that replaces it (Proposition 1).
4. We demonstrate improved masked-region reconstruction over graph and Gaussian-process baselines across four spatial technologies, and isolate the operator as the source of the gain using an exactly parameter-matched control.
5. We establish that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 2), which delimits what any method of this class can identify and predicts the outcome of our perturbation experiment.

2 Method

A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized to $[-1,1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \to \mathbb{R}^C$, factored as encode–evolve–decode.

Spatial encoder. Irregular sampling is handled by kernel-integral layers on a k-NN graph [Li et al., 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j\in\mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement to a per-channel gain, so the layer depends on geometry rather than node identity. Features are splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and an *occupancy* map distinguishing "tissue with no signal" from "no measurement here". Spectral blocks [Li et al., 2021] refine the field.

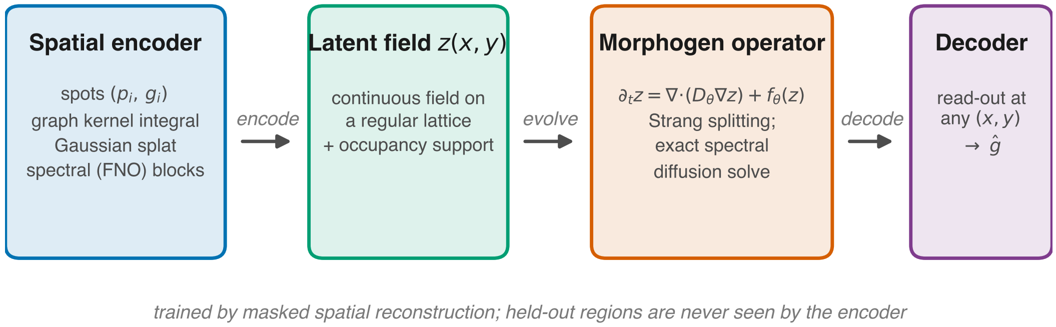

Figure 1: **Neural Morphogen Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

The morphogen operator. The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot \big(\mathbf{D}_\theta \nabla \mathbf{z}\big) + f_\theta(\mathbf{z}), \tag{1}$$

with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as 1×1 convolutions with a $\tanh$-bounded output.

Equation (1) is stiff, with diffusion eigenvalues scaling as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t = O(h^2/\|\mathbf{D}\|)$. For constant-coefficient anisotropic diffusion, however, the operator is diagonal in the Fourier basis and evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp\big(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k}\, \tau\big)\, \mathcal{F}. \tag{2}$$

Composing (2) with a midpoint reaction update under Strang splitting [Strang, 1968] yields the one-step map $S_{\Delta t} = e^{\frac{\Delta t}{2}\mathcal{A}} \circ R_{\Delta t} \circ e^{\frac{\Delta t}{2}\mathcal{A}}$, which is second-order accurate (Proposition **??**). Two structural guarantees follow; both are proved in Appendix **??** and verified numerically in Table **??**.

Because $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2) lies in $(0, 1]$ for *every* Δt: the diffusive half-step is an L^2 contraction and conserves the spatial mean exactly, with no CFL restriction (Proposition **??**). Definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$), so this holds along the entire optimization trajectory — no step size couples to learned parameters and no projection is required — and combining it with $\|f_\theta\|_\infty \leq \|g\|_\infty$ yields a uniform absorbing set for the zero-mean component (Proposition **??**). Section **??** measures what this buys in practice. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths per unit pseudo-time, convertible to microns via the stored coordinate scale.

Decoder. Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, followed by an MLP. *The decoder receives no positional input*: given coordinates it could memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninterpretable.

Objectives. The objective combines reconstruction (MSE plus a per-gene Pearson term, since the spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and physics terms. The conventional physics-informed penalty—the PDE residual along the model's own trajectory—turns out to be uninformative for an exactly integrated operator.

Writing $R_{\Delta t} := \Delta t^{-1}(S_{\Delta t}\mathbf{z} - \mathbf{z}) - (\mathcal{A}_\mathbf{D}\mathbf{z} + f_\theta(\mathbf{z}))$, one finds $\|R_{\Delta t}\| = \frac{\Delta t}{2}\|[\mathcal{A}_\mathbf{D}, \mathcal{F}_\theta]\mathbf{z}\| + O(\Delta t^2)$ uniformly over bounded parameter sets, with no dependence on the observations (Proposition 1). Minimizing $\|R_{\Delta t}\|^2$ therefore cannot fit the model to data—the data do not appear—and merely biases θ toward operators whose diffusive and reactive parts commute (Appendix A). We instead impose the constraint that the *measured configuration be a near-stationary state* of the learned operator,

$$\mathcal{L}_{\mathrm{PDE}} = \left\|\nabla\cdot(\mathbf{D}_\theta\nabla\mathbf{z}_T) + f_\theta(\mathbf{z}_T)\right\|^2, \tag{3}$$

the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$ and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

Interpretability. Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t\delta\mathbf{z}_\mathbf{k} = (\mathbf{J} - \mathbf{k}^\top\mathbf{D}\mathbf{k})\,\delta\mathbf{z}_\mathbf{k}$ with $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max\mathrm{Re}\,\mathrm{spec}(\mathbf{J} - |\mathbf{k}|^2\mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned operator — a falsifiable read-out of the dynamics it has inferred (Appendix **??**).

3 Experimental setup

Data. We use 6 public datasets spanning four spatial technologies (17 tissue sections, 968,794 measured locations; Fig. **??**), plus one non-spatial perturbation screen (111,659 dissociated cells) used only in Section **??**. Raw reads are never used: we consume vendor count matrices and record a SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization, log transform, and HVG selection that *force-includes* every detected morphogen-pathway gene, so the interpretability analysis is not evaluated on a set that selection has already biased. Coordinates are normalized **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned diffusion coefficient meaningless.

Task and splits. Masked spatial reconstruction: contiguous tissue blocks are hidden from the encoder and every model predicts expression at those coordinates. Random spot-level splits leak, since a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

Baselines. A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani et al., 2017], an exact GP and a multi-bandwidth GP [Rasmussen and Williams, 2006], and an implicit neural field with Fourier features and SIREN activations [Tancik et al., 2020]. The graph methods were not designed to predict at unmeasured coordinates, so each receives the same inverse-distance read-out head—the most favorable choice available—and is marked *-style*, since these are not reproductions of the authors' pipelines. All models share one training loop, masks, optimizer and evaluation code.

Metrics. Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al., 2004] on rasterized maps, and absolute error in Moran's I [Moran, 1950], the last included because correlation and RMSE both reward regression toward a smooth mean.

Two scopes, kept distinct. Results are reported at two scales and we label every number with the one it belongs to. The *benchmark* spans 17 sections at 2 seeds under a reduced budget, and is the basis for every claim about generality (Section 4, Table 1); the *primary section* (visium_mouse_brain) is run at the full budget with 3 seeds and carries the diagnostics that need a converged fit — the learned operator, the ablations and the physics read-outs (Appendix **??**, Table **??**). The two are not interchangeable: absolute scores are higher on the primary section, and a value from one is never quoted in support of a claim about the other.

4 Results

4.1 Masked spatial reconstruction across 17 sections

We evaluate on 17 sections spanning four technologies (198 runs), with the section as the unit of analysis: every model sees identical held-out blocks, so a paired test removes the between-section

Table 1: Masked spatial reconstruction across 17 tissue sections spanning four technologies, mean $\pm$ s.d. over sections (2 seeds each, averaged within section). Excluded for incomplete coverage: Gaussian Process (1/17 sections). Stars give Holm-corrected Wilcoxon signed-rank p for the paired comparison against NMO, with the section as the unit of analysis: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Sections on which NMO wins on Pearson r — Autoencoder (non-spatial): 13/17; GNN (GCN): 16/17; GP, multi-bandwidth: 17/17; Neural field (SIREN): 17/17; STAGATE-style: 13/17.

model	n	Pearson r ↑	RMSE ↓	SSIM ↑	$\lvert\Delta I\rvert$ ↓
STAGATE-style	17	$0.154 \pm 0.048^{**}$	$0.704 \pm 0.360^{**}$	$0.306 \pm 0.037^{*}$	$0.766 \pm 0.041^{***}$
GNN (GCN)	17	$0.153 \pm 0.048^{***}$	$0.704 \pm 0.358^{**}$	$0.307 \pm 0.041^{*}$	$0.795 \pm 0.042^{***}$
Autoencoder (non-spatial)	17	$0.153 \pm 0.047^{**}$	$0.728 \pm 0.374^{***}$	$0.224 \pm 0.037^{***}$	$0.756 \pm 0.044^{***}$
GP, multi-bandwidth	17	$0.123 \pm 0.041^{***}$	$0.716 \pm 0.367^{***}$	$0.291 \pm 0.027^{**}$	$0.777 \pm 0.030^{***}$
Neural field (SIREN)	17	$0.106 \pm 0.042^{***}$	$0.725 \pm 0.368^{***}$	$0.235 \pm 0.033^{***}$	$0.625 \pm 0.058^{***}$
NMO (ours)	17	$\mathbf{0.168} \pm \mathbf{0.050}$	$\mathbf{0.697} \pm \mathbf{0.356}$	$\mathbf{0.320} \pm \mathbf{0.042}$	$\mathbf{0.853} \pm \mathbf{0.043}$

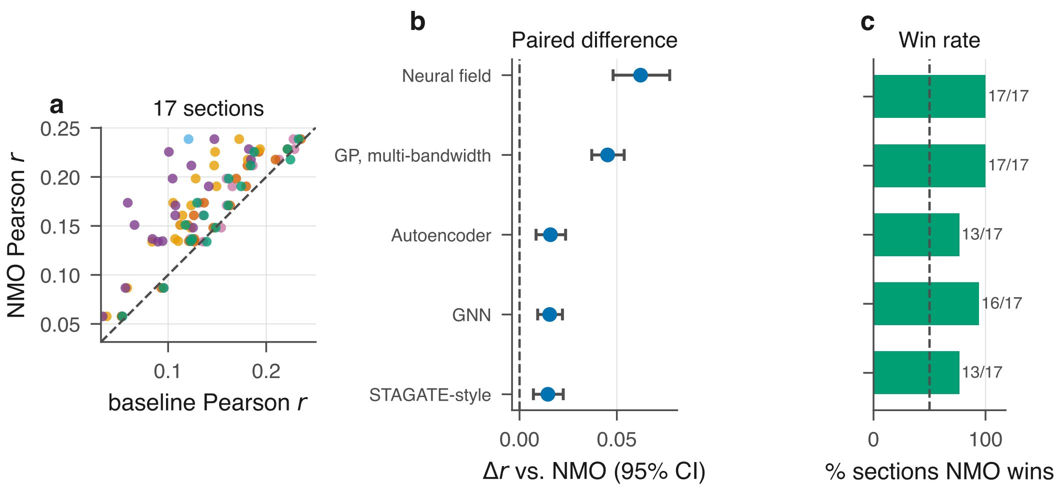

Figure 2: **Benchmark across 17 tissue sections.** (a) NMO against each baseline, one point per section; colors follow the model ordering of (b)–(c). (b) Mean paired difference with a percentile bootstrap 95% CI over sections. (c) Fraction of sections on which NMO wins. Every confidence interval excludes zero.

variance that otherwise dominates. NMO attains Pearson r 0.168 against 0.154 for the closest baseline (STAGATE-style).

All 5 baselines are beaten with Holm-corrected $p \leq 0.003$ (Appendix **??**), and the *smallest* effect size across the family is $d_z = 0.88$, which is large by conventional standards; NMO wins on at least 13 of 17 sections against every comparator. The advantage is therefore consistent across anatomies rather than an artifact of one tissue: the twelve MERFISH sections span the anterior–posterior axis of the same brain with assay, panel and preprocessing held fixed, so the variation between them is anatomical.

The gain comes from continuity, not from message passing. Two controls localize the effect, both measured on this benchmark. A non-spatial autoencoder sees no coordinates and reaches 0.153 purely through the inverse-distance read-out head shared by every graph baseline, so the gap between it and a graph model isolates *message passing*: across 2 graph models that margin spans 0.000 to 0.001 Pearson r, against 0.016 for NMO — an order of magnitude larger. An implicit neural field (SIREN), which is continuous by construction but carries no dynamics, is the weakest model tested (0.106, a gap of 0.062 against NMO). Continuity alone is therefore not sufficient, and on these sections message passing adds little over a well-chosen read-out head; it is the *operator* that carries the gain, which the parameter-matched T=0 control confirms directly (Section 5).

Spatial autocorrelation. NMO leads on SSIM (0.320 against 0.307 for the best baseline, GNN (GCN)) but has by some margin the largest error in Moran's I (0.853 against 0.625 for Neural field (SIREN), a deficit of 0.229): the predicted fields are markedly more spatially autocorrelated than the measurements. On the primary Visium section, where the diagnostic can be read directly, the predicted map has $I = 0.936 \pm 0.004$ against 0.174 measured. SSIM measures local contrast on a rasterized map and Moran's I global autocorrelation on the point set, and the pattern is the expected signature of a dissipative operator attenuating high spatial frequencies — the same diffusion-driven smoothing analyzed for graph neural diffusion [Chamberlain et al., 2021, Choi et al., 2023]. Section 6 discusses the remedy.

4.2 Further results

The exponential integrator costs $32\times$ fewer steps than explicit Euler to reach the same tolerance; the reconstruction does not degrade spatial-domain structure, marker localization or k-NN neighborhoods; zero-shot transfer retains part of the learned structure but is beaten by the STAGATE-style baseline; and a counterfactual perturbation probe returns a null result that Theorem 2 predicts in advance. All are reported in full in the archival version.

5 Ablations

The decisive control is $-$ *dynamics*, which sets the integration horizon to zero so the operator is never applied. The sweep (Appendix **??**) uses a reduced budget at which the full model has not converged (0.233 against 0.247); since undertraining compresses every delta, we re-ran the two decisive variants at the benchmark budget (3 seeds; Appendix **??**), obtaining 0.224 ± 0.005 (-0.023) for T=0 and 0.241 ± 0.003 (-0.006) without the reaction. Disabling the operator therefore costs more than the entire margin over the best baseline.

This control is exactly parameter-matched: the operator holds 9,536 of the model's 2,198,336 parameters (0.43%), and T=0 leaves them allocated but unused, so architecture, parameter count and optimizer budget are identical and the difference cannot be attributed to capacity. This matters because NMO is larger than the graph baselines, and the control separates the two explanations.

6 Limitations

A limitation intrinsic to the setting is that a single stationary snapshot cannot identify a dynamical system. Theorem 2 makes this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^\star(\Omega)$, a Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather than the data. Reported diffusion lengths accordingly characterize the fitted operator rather than physical morphogen transport, and "morphogen" names the operator class—anisotropic reaction–diffusion—rather than a validated claim about signaling. Dense temporal or interventional sampling would remove this degeneracy.

We note that the predicted fields are smoother than the measurements, and that on Moran's I error NMO trails the baselines it leads on the other metrics. This is a known consequence of diffusion-based architectures [Chamberlain et al., 2021, Choi et al., 2023] and bounds the applications we would recommend: prediction at unmeasured coordinates is well supported, delineation of sharp boundaries less so. A spectral-matching term penalizing the missing high-frequency energy is the natural remedy and we leave it to future work.

The benchmark's statistical resolution is limited by the number of independent specimens rather than by the number of sections. Twelve of 17 sections come from one brain, leaving 5 independent specimens, at which the smallest attainable Wilcoxon p is 0.06. Against the continuous baselines this is enough: NMO wins on every specimen, which is the strongest result the design admits. Against the graph baselines it is not ($d_z = 0.32 - -0.81$, NMO ahead on 3 of 5); the section-level separation there rests substantially on correlated sections of one brain, and additional independent tissues —

not additional sections — are what would settle it. We regard this as the single most informative experiment left undone.

Transfer remains the weakest setting. NMO retains a substantial fraction of in-domain performance zero-shot, but the STAGATE-style baseline transfers better, and until $\mathbf{D}_\theta$ is conditioned on tissue-level covariates the operator is best regarded as a within-section model. Finally, the *-style* baselines implement each method's architectural core with a common read-out head and are not reproductions of published pipelines; the benchmark uses a reduced epoch budget and subsampled sections so that 198 runs were feasible on CPU, which lowers all models' absolute scores; and scope is limited to single 2-D sections.

References

Benjamin Paul Chamberlain, James Rowbottom, Maria Gorinova, Stefan Webb, Emanuele Rossi, and Michael M. Bronstein. GRAND: Graph neural diffusion. In *International Conference on Machine Learning (ICML)*, 2021.

Kok Hao Chen, Alistair N. Boettiger, Jeffrey R. Moffitt, Siyuan Wang, and Xiaowei Zhuang. Spatially resolved, highly multiplexed RNA profiling in single cells. *Science*, 348(6233):aaa6090, 2015.

Jeongwhan Choi, Seoyoung Hong, Noseong Park, and Sung-Bae Cho. GREAD: Graph neural reaction diffusion networks. In *International Conference on Machine Learning (ICML)*, 2023.

Kangning Dong and Shihua Zhang. Deciphering spatial domains from spatially resolved transcriptomics with an adaptive graph attention auto-encoder. *Nature Communications*, 13(1):1739, 2022.

Alfred Gierer and Hans Meinhardt. A theory of biological pattern formation. *Kybernetik*, 12(1): 30–39, 1972.

Jian Hu, Xiangjie Li, Kyle Coleman, Amelia Schroeder, Nan Ma, David J. Irwin, Edward B. Lee, Russell T. Shinohara, and Mingyao Li. SpaGCN: Integrating gene expression, spatial location and histology to identify spatial domains and spatially variable genes by graph convolutional network. *Nature Methods*, 18(11):1342–1351, 2021.

Anna Kicheva, Periklis Pantazis, Tobias Bollenbach, Yannis Kalaidzidis, Thomas Bittig, Frank Jülicher, and Marcos González-Gaitán. Kinetics of morphogen gradient formation. *Science*, 315 (5811):521–525, 2007.

Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations (ICLR)*, 2017.

Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Graph kernel network for partial differential equations. *arXiv preprint arXiv:2003.03485*, 2020.

Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations (ICLR)*, 2021.

Patrick A. P. Moran. Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2):17–23, 1950.

Patrick Müller, Katherine W. Rogers, Ben M. Jordan, Joon S. Lee, Drew Robson, Sharad Ramanathan, and Alexander F. Schier. Differential diffusivity of Nodal and Lefty underlies a reaction-diffusion patterning system. *Science*, 336(6082):721–724, 2012.

Giovanni Palla, Hannah Spitzer, Michal Klein, David Fischer, Anna Christina Schaar, Louis Benedikt Kuemmerle, Sergei Rybakov, et al. Squidpy: A scalable framework for spatial omics analysis. *Nature Methods*, 19(2):171–178, 2022.

Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.

Rushikesh Sheth, Luciano Marcon, M. Félix Bastida, Marisa Junco, Laura Quintana, Randall Dahn, Marie Kmita, James Sharpe, and Maria A. Ros. Hox genes regulate digit patterning by controlling the wavelength of a Turing-type mechanism. *Science*, 338(6113):1476–1480, 2012.

Patrik L. Ståhl, Fredrik Salmén, Sanja Vickovic, Anna Lundmark, José Fernández Navarro, Jens Magnusson, Stefania Giacomello, et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science*, 353(6294):78–82, 2016.

Gilbert Strang. On the construction and comparison of difference schemes. *SIAM Journal on Numerical Analysis*, 5(3):506–517, 1968.

Matthew Tancik, Pratul P. Srinivasan, Ben Mildenhall, Sara Fridovich-Keil, Nithin Raghavan, Utkarsh Singhal, Ravi Ramamoorthi, Jonathan T. Barron, and Ren Ng. Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Alan M. Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London B*, 237(641):37–72, 1952.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.

Lewis Wolpert. Positional information and the spatial pattern of cellular differentiation. *Journal of Theoretical Biology*, 25(1):1–47, 1969.

Shuizi Rachel Yu, Markus Burkhardt, Matthias Nowak, Jonas Ries, Zdeněk Petrášek, Steffen Scholpp, Petra Schwille, and Michael Brand. Fgf8 morphogen gradient forms by a source-sink mechanism with freely diffusing molecules. *Nature*, 461(7263):533–536, 2009.

A Vacuity of the along-trajectory residual

Physics-informed training conventionally penalizes the PDE residual evaluated on the model's own trajectory. The following makes precise why that penalty is uninformative for the present architecture, and motivates the steady-state formulation adopted instead.

Proposition 1 (The trajectory residual is an integrator diagnostic)**.** *Define* $R_{\Delta t}(\mathbf{z};\theta) := \Delta t^{-1}\big(S_{\Delta t}\mathbf{z}-\mathbf{z}\big)-\big(\mathcal{A}_{\mathbf{D}}\mathbf{z}+f_\theta(\mathbf{z})\big)$*. Then for every* θ *in a bounded parameter set and every sufficiently regular* $\mathbf{z}$*,*

$$\|R_{\Delta t}(\mathbf{z};\theta)\| \;=\; \Delta t\cdot\left\|\tfrac{1}{2}\big[\mathcal{A}_{\mathbf{D}},\,\mathcal{F}_\theta\big]\mathbf{z}\right\| \;+\; O(\Delta t^2), \tag{4}$$

where $\mathcal{F}_\theta\mathbf{z} := f_\theta(\mathbf{z})$ *and* $[\cdot,\cdot]$ *is the commutator. In particular* $R_{\Delta t}\to 0$ *as* $\Delta t\to 0$ uniformly in θ*, and* $R_{\Delta t}$ *does not depend on the observed data.*

Proof sketch. Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading commutator term at order Δt; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are evaluated at the same state $\mathbf{z}$. □

Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the observations do not appear in (4); it drives θ toward operators whose diffusive and reactive parts commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$ and therefore vanishes in the very limit in which the discretization is most faithful. The empirical scaling is confirmed in Table **??** (observed order 1.00).

The constraint retained instead is the *steady-state* residual (3), which asks that the measured configuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it constrains $\mathbf{D}_\theta$ and f_θ jointly: the set of pairs satisfying it is a proper subset of parameter space, characterized next.

B Non-identifiability from a single snapshot

The following records the central epistemic limitation of the setting: an operator fitted to one stationary field is determined only on a measure-zero subset of its own domain. The argument is elementary — a C^1 image of a two-dimensional domain cannot fill $\mathbb{R}^C$ once $C\geq 3$, so the stationarity constraint simply says nothing off that image — and we state it as a theorem for its consequence rather than for its difficulty. That consequence is sharp and is not, in our reading, generally acknowledged in this literature: it means every counterfactual an operator of this class produces is a statement about the architecture, not about the data, and it predicts in advance the null result of Section **??**.

Theorem 2 (Latent-space non-identifiability)**.** *Let* $\mathbf{z}^\star\in C^2(\Omega;\mathbb{R}^C)$ *with* $\Omega\subset\mathbb{R}^2$ *open, and suppose* $C\geq 3$*. Fix any* $\mathbf{D}\in(\mathcal{S}^2_{++})^C$ *and define the admissible set*

$$\mathcal{F}(\mathbf{D},\mathbf{z}^\star) := \Big\{\, f\in C^1(\mathbb{R}^C;\mathbb{R}^C) \;:\; \mathcal{A}_{\mathbf{D}}\mathbf{z}^\star(x)+f\big(\mathbf{z}^\star(x)\big)=0 \;\;\forall x\in\Omega \,\Big\}. \tag{5}$$

If $\mathcal{F}(\mathbf{D},\mathbf{z}^\star)\neq\emptyset$ *then it is an infinite-dimensional affine subset of* $C^1(\mathbb{R}^C;\mathbb{R}^C)$*. Specifically, for every* $\varphi\in C^1(\mathbb{R}^C;\mathbb{R}^C)$ *vanishing on* $\mathcal{R}:=\mathbf{z}^\star(\Omega)$ *and every* $f\in\mathcal{F}$*, one has* $f+\varphi\in\mathcal{F}$*, and the space of such* φ *is infinite-dimensional.*

Proof. The constraint restricts f only through its values on $\mathcal{R}=\mathbf{z}^\star(\Omega)$; hence $f+\varphi$ satisfies it whenever $\varphi|_{\mathcal{R}}\equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^\star$ is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C\geq 3$, $\dim_H(\mathcal{R})\leq 2<C$, so $\mathcal{R}$ is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B; choosing countably many disjoint balls $B_i\subset B$ and bump functions $\varphi_i\in C_c^\infty(B_i;\mathbb{R}^C)$ yields a linearly independent family, all vanishing on $\mathcal{R}$. □

Corollary 3 (What a stationarity fit can determine)**.** *Under the hypotheses of Theorem 2, the steady-state loss* (3) *constrains f_θ only on $\mathcal{R}$, a set of Lebesgue measure zero in $\mathbb{R}^C$. Any statement about f_θ off $\mathcal{R}$—in particular any counterfactual obtained by displacing the latent state away from the observed manifold—is determined by the inductive bias of the architecture and the regularizers, not by the data.*

Corollary 3 has a direct experimental consequence. The in-silico perturbation of Section **??** displaces $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states off $\mathcal{R}$, where Theorem 2 shows the data exert no constraint. The null outcome reported there is therefore consistent with the theory rather than merely an empirical disappointment: a single stationary snapshot does not contain the information such a counterfactual would require. Breaking the degeneracy requires observations that visit a C-dimensional neighborhood of $\mathcal{R}$—dense temporal sampling, or interventional data.